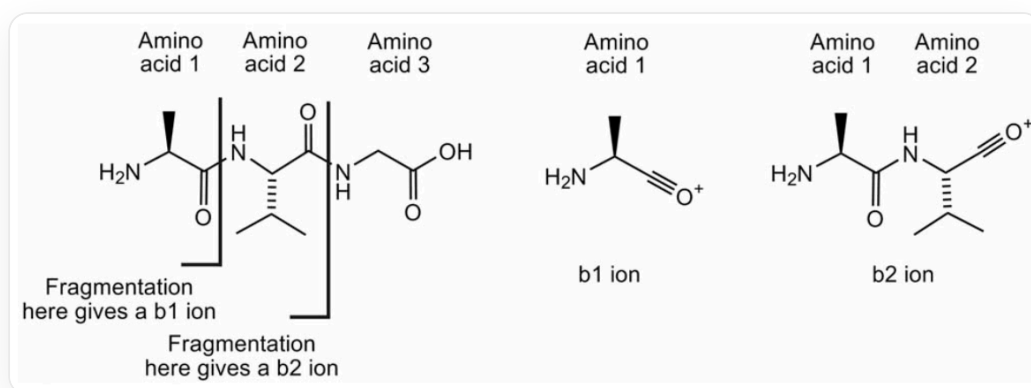
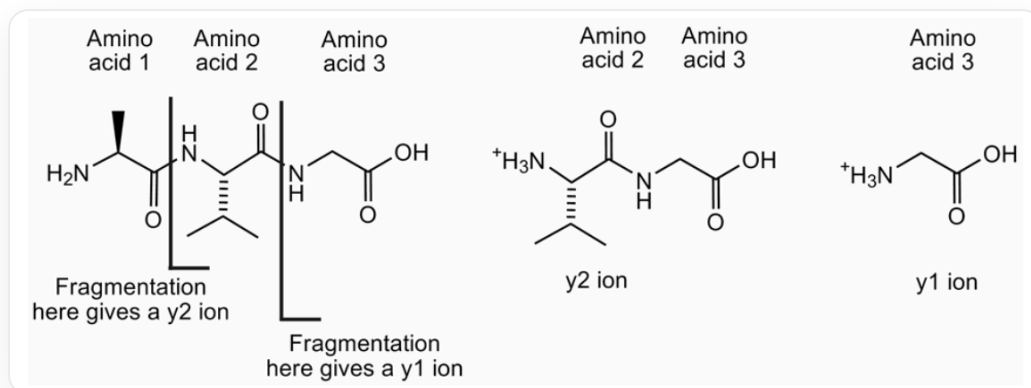


Question

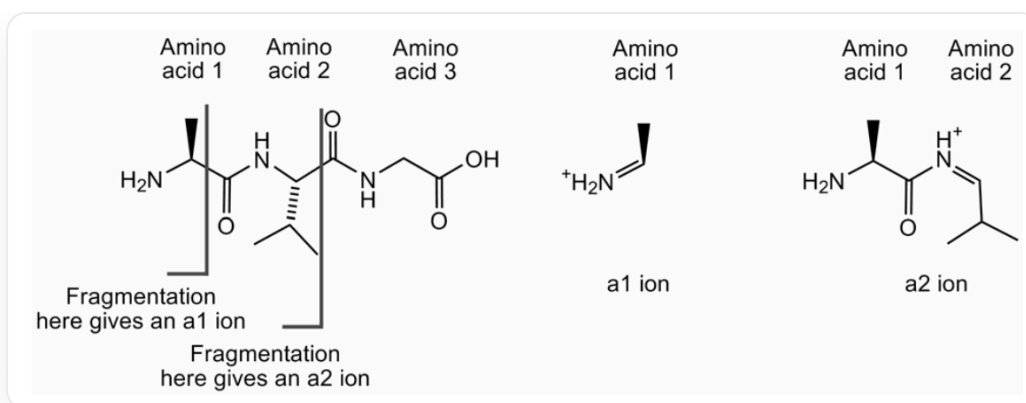
Tandem mass spectrometry is a rapid method for peptide sequencing. This technique involves the formation of precursor ions and their fragmentation into smaller ions. In peptides, fragmentation often occurs on the peptide backbone. The naming of fragment ions depends on the site of fragmentation and which atom carries the positive charge. For example, some ions formed from alanyl-leucyl-glycine are shown below:



For the tripeptide alanyl-leucyl-glycine, cleavage of the amide bond between alanine and leucine yields an N-terminal fragment, the alanyl ion, denoted as b_1 ion. Cleavage of the amide bond between leucine and glycine yields an N-terminal fragment, the alanyl-leucyl ion, denoted as b_2 ion



For the tripeptide alanyl-leucyl-glycine, cleavage of the amide bond between alanine and leucine yields a C-terminal fragment, the protonated leucyl-glycine ammonium ion, denoted as y_2 ion. Cleavage of the amide bond between leucine and glycine yields a C-terminal fragment, the protonated glycine ammonium ion, denoted as y_1 ion

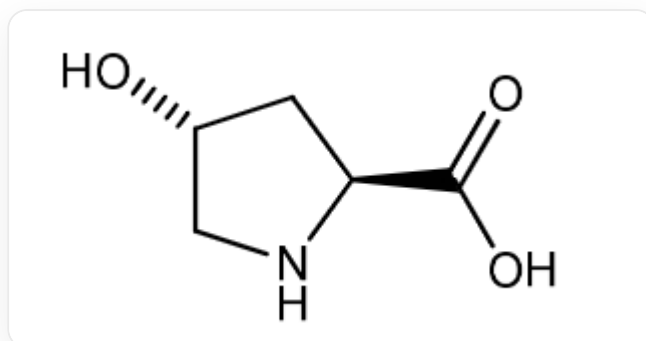


For the tripeptide alanyl-leucyl-glycine, cleavage of the carbon-carbon bond between the carbonyl carbon and the carbonyl α -carbon within alanine yields an N-terminal fragment, the protonated immonium ion, denoted as a_1 ion. Cleavage of the carbon-carbon bond between the carbonyl carbon and the carbonyl α -carbon within leucine-glycine yields an N-terminal fragment, the protonated alanyl immonium ion, denoted as a_2 ion

Scientists extracted a protein from a fossilized bone of a prehistoric organism. The mass spectrometry data of a 19-peptide fragment is shown in the following table:

ion	m/z	ion	m/z	ion	m/z	ion	m/z
y_1	175.1	b_5	715.3	y_8	986.5	b_{12}	1400.7
a_2	249.1	y_6	726.4	b_9	1069.5	y_{14}	1508.8
y_2	272.2	a_6	800.4	y_9	1083.5	b_{14}	1612.7
y_3	401.2	y_7	823.4	b_{10}	1140.5	a_{15}	1681.8
a_4	501.2	b_6	828.4	a_{11}	1209.6	y_{15}	1694.9
b_4	529.2	b_7	885.4	y_{11}	1267.6	y_{16}	1831.9
y_5	611.4	a_8	928.4	y_{12}	1338.7	y_{17}	1946.9
a_5	687.3	b_8	956.5	y_{13}	1395.7	b_{17}	1951.9

It is known that the first two amino acids of the peptide are Tyr-Leu, and the peptide sequence includes a hydroxyproline, represented as Hyp, with a mass of 131.1, and the structure is as follows:



Using the mass spectrometry ion mass table, determine the most accurate possible peptide sequence. In the peptide sequence, for amino acids whose exact type cannot be determined, consider all possibilities for multiple amino acids present at a position, based on calculations and reasoning.

Based on your inferred most likely peptide sequence, select the closest modern species from the partial sequences of homologous proteins obtained from the following modern species (in the following sequences, hydroxyproline and proline are both abbreviated as P):

Species	Sequence
Carp	DLTVAQLESLKEVCEANLACEHMMDVSGIIAAYTAYYGPIPY
Chicken	HYAQDSGVAGAPPNPLEAQREVCELSPDCDELADQIGFQEAYRRFYGPV
Cow	YLDHWLGAPAPYPDPLEPKREVCELNPDCDELADHIGFQEAYRRFYGPV
Horse	YLDHWLGAPAPYPDPLEPRREVCELNPDCDELADHIGFQEAYRRFYGPV
Human	YLYQWLGAPVPYPDPLEPRREVCELNPDCDELADHIGFQEAYRRFYGPV
Rabbit	QLINGQGAPAPYPDPLEPKREVCELNPDCDELADQVGLQDAYQRFYGPV
Sheep	YLDPGLGAPAPYPDPLEPRREVCELNPDCDELADHIGFQEAYRRFYGPV
Toad	SYGNNVGQGAAVGSPLESQREVCELNPDCDELADHIGFQEAYRRFYGPV

A. Carp

B. Chicken

C. Cow

D. Horse

E. Human

F. Rabbit

G. Sheep

H. Toad

Answer

Correct Answer: **D**

Detailed Explanation

The mass of the ion y_1 can be used to determine the identity of the last amino acid in a polypeptide. The mass of the y_1 ion is one mass unit greater than the corresponding amino acid; therefore, the last amino acid must be arginine (Arg).

CHECKPOINT

1 PTS

Indicates that y_1 is arginine Arg

The y-series ions are the most complete, and the sequence can be determined by comparing the masses of successive y ions:

Ion	m/z	Mass difference between b_n and b_{n-1}	Corresponding amino acid	Amino acid mass
y_1	175.1			
y_2	272.2	97.1	18	115.1
y_3	401.2	129.0	17	147.0
y_4				
y_5	611.4			
y_6	726.4	115.0	14	133.0
y_7	823.4	97.1	13	115.1
y_8	986.5	163.1	12	181.1
y_9	1083.5	97.1	11	115.1
y_{10}				
y_{11}	1267.6			
y_{12}	1338.7	71.0	8	89.0
y_{13}	1395.7	57.0	7	75.0
y_{14}	1508.8	113.1	6	131.1
y_{15}	1694.9	186.1	5	204.1

Ion	m/z	Mass difference between b_n and b_{n-1}	Corresponding amino acid	Amino acid mass
y_{16}	1831.9	137.1	4	155.1
y_{17}	1946.9	115.0	3	133.0

CHECKPOINT

1 PTS

Analyze the sequence of y-series ions

Based on the y-series, the sequence is:

Tyr-Leu-Asp-His-Trp-Leu/Ile/Hyp-Gly-Ala-xxx-xxx-Pro-Tyr-Pro-Asp-xxx-xxx-Glu-Pro-Arg

CHECKPOINT

5 PTS

The polypeptide sequence obtained from the y-series ions is Tyr-Leu-Asp-His-Trp-Leu/Ile/Hyp-Gly-Ala-[undetermined]-[undetermined]-Pro-Tyr-Pro-Asp-[undetermined]-[undetermined]-Glu-Pro-Arg

The identity of the 15th amino acid in the sequence can be determined by the mass difference between b_{14} and a_{15} :

$$M_r(\text{amino acid 15}) = \text{mass}(a_{15}) - \text{mass}(a_{14}) + M_r(\text{C}) + 2M_r(\text{O}) + 2M_r(\text{H}) = 115.0$$

Therefore, amino acid 15 must be proline (Pro).

CHECKPOINT

1 PTS

Determine that amino acid number 15 is proline Pro

The mass difference between ions y_3 and y_5 gives the fragment mass corresponding to amino acids 15 and 16.

$$M_r(\text{15-16 dipeptide}) = \text{mass}(y_5) - \text{mass}(y_3) + M_r(H_2O)$$

$$M_r(\text{amino acid 16}) = M_r(\text{15-16 dipeptide}) - M_r(\text{amino acid 15}) + M_r(H_2O) = 131.1$$

Therefore, amino acid 16 must be isoleucine (Ile), leucine (Leu), or hydroxyproline (Hyp).

CHECKPOINT

1 PTS

Determine that amino acid number 16 is isoleucine Ile, leucine Leu, or hydroxyproline Hyp

The mass of amino acid 10 can be determined by the mass difference between b_9 and b_{10} :

$$M_r(\text{amino acid 10}) = \text{mass}(b_{10}) - \text{mass}(b_9) + M_r(H_2O)$$

Amino acid 10 is alanine (Ala).

CHECKPOINT

1 PTS

Determine that amino acid number 10 is alanine Ala

The mass difference between ions y_{11} and y_9 gives the fragment mass corresponding to amino acids 9 and 10.

$$M_r(9\text{-}10 \text{ dipeptide}) = \text{mass}(y_{11}) - \text{mass}(y_9) + M_r(H_2O)$$

$$M_r(\text{amino acid } 9) = M_r(9\text{-}10 \text{ dipeptide}) - M_r(\text{amino acid } 10) + M_r(H_2O)$$

The mass of amino acid 9 is 131.1, so it must be isoleucine (Ile), leucine (Leu), or hydroxyproline (Hyp).

CHECKPOINT

1 PTS

Determine that amino acid number 9 is isoleucine Ile, leucine Leu, or hydroxyproline Hyp

Therefore, the sequence of the polypeptide is:

Tyr-Leu-Asp-His-Trp-Leu/Ile/Hyp-Gly-Ala-Leu/Ile/Hyp-Ala-Pro-Tyr-Pro-Asp-Pro-Leu/Ile/Hyp-Glu-Pro-Arg

In order to verify the relationship between the prehistoric protein A and modern species, the inferred 19-peptide sequence was aligned with the homologous protein sequences of various modern organisms given. The purpose of the alignment is to find the sequence with the highest degree of matching.

First, represent the ancient protein sequence finally inferred from the question. This sequence contains three uncertain amino acid sites (positions 6, 9, 16), which may be leucine (L), isoleucine (I), or hydroxyproline (Hyp).

According to the information in the question, hydroxyproline (Hyp) and proline (P) are both represented by **P** in modern species sequences. Therefore, in the alignment, these three ambiguous sites **X** can match **L**, **I**, or **P**.

Ancient 19-peptide sequence pattern:

`Y-L-D-H-W-`X`-G-A-`X`-A-P-Y-P-D-P-`X`-E-P-R`

The above ancient sequence pattern is aligned one by one with the first 19 amino acids of the homologous protein sequence of each modern species, and the number of matched amino acids is calculated. Mismatched amino acids are indicated in bold.

Species	Alignment sequence (first 19 amino acids)	Number of mismatches	Similarity
Ancient sequence pattern	`Y L D H W X G A X A P Y P D P X E P R`	-	-
Horse	`Y L D H W L G A P A P Y P D P L E P R`	0	19/19
Cow	`Y L D H W L G A P A P Y P D P L E P` `K`	1	18/19
Sheep	`Y L D` `P` `G` `L` `G A P A P Y P D P L E P R`	2	17/19
Human	`Y L` `Y` `Q` `W L G A P` `V` `P Y P D P L E P R`	3	16/19
Rabbit	`Q` `L` `I` `N` `G` `Q` `G A P A P Y P D P L E P` `K`	6	13/19
Chicken	`H` `Y` `A` `Q` `D` `S` `G` `V` `A` `G` `A` `P` `P` `N` `P L E` `A` `Q`	14	5/19
Toad	`S` `Y` `G` `N` `N` `V` `G` `Q` `G` `A` `A` `V` `G` `S` `P L E` `S` `Q`	15	4/19
Carp	`D` `L` `T` `V` `A` `Q` `L` `E` `S` `L` `K` `E` `V` `C` `E` `A` `N` `L` `A`	16	3/19

CHECKPOINT

3 PTS

Perform sequence alignment and obtain the correct results, the order of polypeptide sequence similarity is horse>cow>sheep>human>rabbit>chicken>toad>carp

From the alignment results in the table above, it can be seen that the sequence is most similar to that of the horse.

CHECKPOINT

1 PTS

Conclude that the most similar sequence is that of the horse